COMPUTER FUNDAMENTAL LAB REPORT # 5

NAME	**Qazi Ehsanullah**
REGISTRATION NUMBER	**BF25NWELE0738**
DEPARTMENT	**ELECTRICAL ENG (Power)**
SECTION	**"A"**
SEMISTER	**2nd**
SUBJECT	**COMPUTER PROGRAMING**

1. Objective:

The purpose of this lab is to learn and apply three types of loops in C++:

- ***The while loop***:

  Executing code based on a pre-checked condition.

- ***The nested while loop***:

  Running a loop inside another loop for complex iterations.

- ***The do...while loop:***

Ensuring a block of code runs at least once before checking the condition.

2. Methodology & Examples

The while Loop

A "pre-test" loop. If the condition is false at the start, the code block is never executed.

- ***Example:***

  Printing "Hello" 3 times.

```
#include <iostream>
using namespace std;
int main()
    {

        int count = 0;
        while(count < 3)
 {

    cout << "Hello" << endl;
    count++;
}



return 0;
}
```

The do...while Loop

A "post-test" loop. This is useful for user menus where you want the prompt to show up at least once regardless of the condition.

- ***Example:***

```
#include <iostream>
using namespace std;
int main()
    {
    int x = 10;
do {
    cout << x; // Executes once even though 10 is not < 5
} while (x < 5);



return 0;
}
```

nested while loop :

A nested while loop is a programming structure where one while loop is placed inside the body of another while loop.

EXAMPLE:

```
#include <iostream>
using namespace std;

int main() {
    int i = 1; // Outer loop counter (Rows)
    while (i <= 5) {
        int j = 1; // Inner loop counter (Columns) - RE-INITIALIZED HERE
        while (j <= 5) {
            cout << (i * j) << "\t"; // Multiplication result
            j++; // Increment inner counter
        }
        cout << endl; // Move to next row after inner loop finishes
        i++; // Increment outer counter
    }
    return 0;
}
```